

Hospital Management Appointment System

Project Members:

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Introduction

Health care has always been one of the most data intensive domains, where the management of patient, doctor, and appointment information can directly impact the quality of care delivered. Despite of technology, some hospitals and clinics in Pakistan maintain paper based records, which causes data inconsistency, and the data can be easily lost in case of any disaster. The focus of this project is to address this loophole.

This project focuses on appointment workflow for doctors, administrators, and staff. Patients can easily book appointments of the designated doctors. Additionally, a doctor can review his day's appointments.

Background

Before choosing the Hospital Appointment Management System, the group considered several options including the University Event Management System but that seemed to be a little irrelevant against the Hospital Appointment Management System, as the healthcare department has more real world relevance.

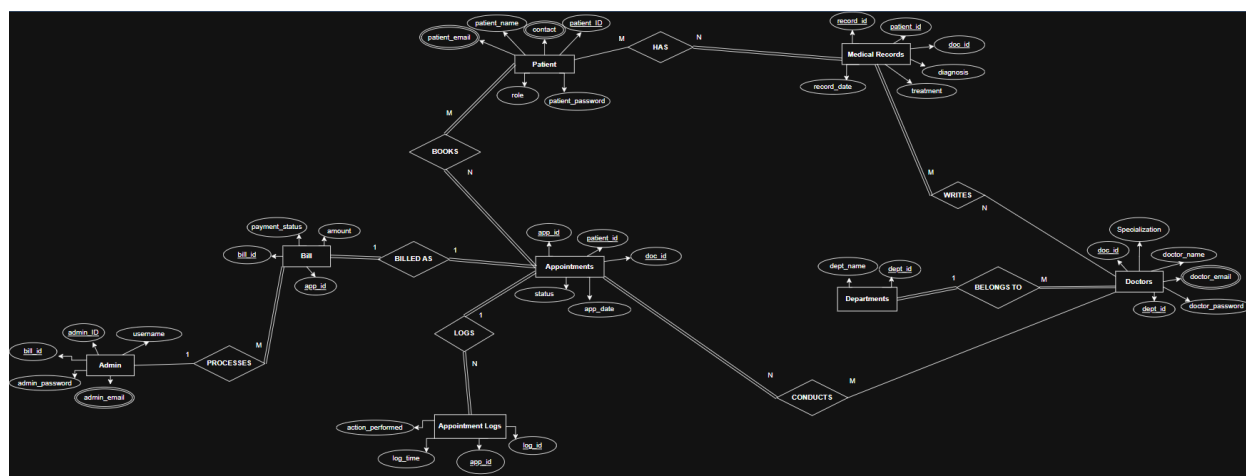
Before building our own the group closely analyzed the current hospital management systems working in Pakistan, so that we have an idea of the functionalities. This really helped us to add some of the features to our project.

Project Specification

User Roles

- ❖ **Patient:** register an account, log in, book appointments, view appointment history
- ❖ **Doctor:** register an account, log in, view scheduled appointments, view appointment history
- ❖ **Administrator:** register an account, log in, view scheduled appointments, update billing status

ERD Diagram



Normalization

Unnormalized table

A	B	C	D	E	F	G	H	I	J
dept_id	dept_name	doc_id	doctor_name	doctor_email	doctor_password	specialization	patient_id	patient_name	patient_email
1001	Cardiology	2001	Dr. Safa Rehman	safa@gmail.com	safa123	Cardiologist	3001	Safwan Ali	safwan@gmail.com
1002	Neurology	2002	Dr. Khadijah Rasheed	khadijah@gmail.com	khadijah123	Neurosurgeon	3002	Amna Rizvi	amnarizvi@example.com, amnarizvi88@example.com
1003	Psychiatrist	2003	Dr. Mohammad Nafees	nafees@gmail.com, nafees90@gmail.com	nafees123	Psychiatrist	3003	Nabiha Ashfaq	nabihaashfaq@example.com
1004	General Physician	2004	Dr. Mahnoor Mansoor	mahnoor@gmail.com	mahnoor123	General Physician	3004	Laksh Kumar	laksh@example.com, lakshi89@example.com
1005	Nephrologist	2005	Dr. Maheen Haseeb	maheen@gmail.com	maheen123	Nephrologist	3005	Haider Ali	haider@example.com

K	L	M	N	O	P	Q	R	S	T	U
patient_password	contact	role	admin_id	username	admin_email	admin_password	app_id	app_date	status	record_id
murakam129	3001234567, 034572789	patient	4001	Abdullah Sheikh	abd@gmail.com	abd1234	5001	5/10/2026	Scheduled	6001
yhabibiuni	300122687, 0436788920	patient	4002	Rehan Asghar	rehan@gmail.com	rehan1234	5002	5/12/2026	Scheduled	6002
polisc123	3007784967	patient	4003	Talha Ahmed	talha@gmail.com	talha1234	5003	5/15/2026	Completed	6003
laksh18	3002188840, 053427898	patient	4004	Rehan Asghar	rehan@gmail.com	rehan1234	5004	5/20/2026	Scheduled, Completed	6004
gik28	3001918781	patient	4005	Talha Ahmed	talha@gmail.com	talha1234	5005	5/6/2026	Completed	6005

U	V	W	X	Y	Z	AA	AB	AC	AD
record_id	diagnosis	treatment	record_date	log_id	action_performed	log_time	bill_id	amount	payment_status
6001	Seasonal Flu	Paracetamol	3/2/2026	7001	Created	5/2/2026 10:30	8001	1500	Unpaid
6002	Cholesterol	Lesser salt intake	2/1/2026	7002	Updated	5/2/2026 14:15	8002	1500	Unpaid
6003	Ulcer	Biopsy	8/1/2026	7003	Canceled	5/3/2026 9:00	8003	1500	Paid
6004	Cholesterol	Lesser salt intake	3/4/2026	7004	Created	12/25/2026 18:00	8004	1500	Unpaid
6005	Seasonal Flu	Paracetamol, Panadol	8/6/2026, 1/1/2026	7005	Updated	1/1/2026 0:01	8005	1500	Paid

1NF form

A	B	C	D	E	F	G	H	I	J
dept_id	dept_name	doc_id	doctor_name	doctor_email	doctor_password	specialization	patient_id	patient_name	patient_email
1001	Cardiology	2001	Dr. Safa Rehman	safa@gmail.com	safa123	Cardiologist	3001	Safwan Alvi	safwan@gmail.com
1001	Cardiology	2001	Dr. Safa Rehman	safa@gmail.com	safa123	Cardiologist	3001	Safwan Alvi	safwan@gmail.com
1002	Neurology	2002	Dr. Khadijah Rasheed	khadijah@gmail.com	khadijah123	Neurosurgeon	3002	Amna Rizvi	amnarizvi@example.com
1002	Neurology	2002	Dr. Khadijah Rasheed	khadijah@gmail.com	khadijah123	Neurosurgeon	3002	Amna Rizvi	amnarizvi@example.com
1002	Neurology	2002	Dr. Khadijah Rasheed	khadijah@gmail.com	khadijah123	Neurosurgeon	3002	Amna Rizvi	amnarizvi88@example.com
1002	Neurology	2002	Dr. Khadijah Rasheed	khadijah@gmail.com	khadijah123	Neurosurgeon	3002	Amna Rizvi	amnarizvi88@example.com
1003	Psychiatrist	2003	Dr. Mohammad Nafees	nafees@gmail.com	nafees123	Psychiatrist	3003	Nabha Ashfaq	nabhaashfaq@example.com
1003	Psychiatrist	2003	Dr. Mohammad Nafees	nafees90@gmail.com	nafees123	Psychiatrist	3003	Nabha Ashfaq	nabhaashfaq@example.com
1004	General Physician	2004	Dr. Mahnoor Mansoor	mahnoor@gmail.com	mahnoor123	General Physician	3004	Laksh Kumar	laksh@example.com
1004	General Physician	2004	Dr. Mahnoor Mansoor	mahnoor@gmail.com	mahnoor123	General Physician	3004	Laksh Kumar	laksh@example.com
1004	General Physician	2004	Dr. Mahnoor Mansoor	mahnoor@gmail.com	mahnoor123	General Physician	3004	Laksh Kumar	laksh@example.com
1004	General Physician	2004	Dr. Mahnoor Mansoor	mahnoor@gmail.com	mahnoor123	General Physician	3004	Laksh Kumar	laksh@example.com
1004	General Physician	2004	Dr. Mahnoor Mansoor	mahnoor@gmail.com	mahnoor123	General Physician	3004	Laksh Kumar	laksh89@example.com
1004	General Physician	2004	Dr. Mahnoor Mansoor	mahnoor@gmail.com	mahnoor123	General Physician	3004	Laksh Kumar	laksh89@example.com
1004	General Physician	2004	Dr. Mahnoor Mansoor	mahnoor@gmail.com	mahnoor123	General Physician	3004	Laksh Kumar	laksh89@example.com
1004	General Physician	2004	Dr. Mahnoor Mansoor	mahnoor@gmail.com	mahnoor123	General Physician	3004	Laksh Kumar	laksh89@example.com
1005	Nephrologist	2005	Dr. Maheen Haseeb	maheen@gmail.com	maheen123	Nephrologist	3005	Haider Ali	haider@example.com
1005	Nephrologist	2005	Dr. Maheen Haseeb	maheen@gmail.com	maheen123	Nephrologist	3005	Haider Ali	haider@example.com
1005	Nephrologist	2005	Dr. Maheen Haseeb	maheen@gmail.com	maheen123	Nephrologist	3005	Haider Ali	haider@example.com
1005	Nephrologist	2005	Dr. Maheen Haseeb	maheen@gmail.com	maheen123	Nephrologist	3005	Haider Ali	haider@example.com

K	L	M	N	O	P	Q	R	S	T	U	V
patient_password	contact	role	admin_id	username	admin_email	admin_password	app_id	app_date	status	record_id	diagnosis
murakami29	3001234567	patient	4001	Abdullah Sheikh	abd@gmail.com	abd1234	5001	2026-05-10 00:00:00	Scheduled	8001	Seasonal Flu
murakami29	034572789	patient	4001	Abdullah Sheikh	abd@gmail.com	abd1234	5001	2026-05-10 00:00:00	Scheduled	8001	Seasonal Flu
habbiuni	300122697	patient	4002	Rehan Asghar	rehan@gmail.com	rehan1234	5002	2026-05-12 00:00:00	Scheduled	8002	Cholesterol
habbiuni	043678920	patient	4002	Rehan Asghar	rehan@gmail.com	rehan1234	5002	2026-05-12 00:00:00	Scheduled	8002	Cholesterol
habbiuni	300122697	patient	4002	Rehan Asghar	rehan@gmail.com	rehan1234	5002	2026-05-12 00:00:00	Scheduled	8002	Cholesterol
habbiuni	043678920	patient	4002	Rehan Asghar	rehan@gmail.com	rehan1234	5002	2026-05-12 00:00:00	Scheduled	8002	Cholesterol
poliscic23	3007784967	patient	4003	Talha Ahmed	talha@gmail.com	talha1234	5003	2026-05-15 00:00:00	Completed	8003	Ulcer
poliscic23	3007784967	patient	4003	Talha Ahmed	talha@gmail.com	talha1234	5003	2026-05-15 00:00:00	Completed	8003	Ulcer
laksh18	3002188840	patient	4004	Rehan Asghar	rehan@gmail.com	rehan1234	5004	2026-05-20 00:00:00	Scheduled	8004	Cholesterol
laksh18	3002188840	patient	4004	Rehan Asghar	rehan@gmail.com	rehan1234	5004	2026-05-20 00:00:00	Completed	8004	Cholesterol
laksh18	053427898	patient	4004	Rehan Asghar	rehan@gmail.com	rehan1234	5004	2026-05-20 00:00:00	Scheduled	8004	Cholesterol
laksh18	053427898	patient	4004	Rehan Asghar	rehan@gmail.com	rehan1234	5004	2026-05-20 00:00:00	Completed	8004	Cholesterol
laksh18	3002188840	patient	4004	Rehan Asghar	rehan@gmail.com	rehan1234	5004	2026-05-20 00:00:00	Scheduled	8004	Cholesterol
laksh18	3002188840	patient	4004	Rehan Asghar	rehan@gmail.com	rehan1234	5004	2026-05-20 00:00:00	Completed	8004	Cholesterol
laksh18	053427898	patient	4004	Rehan Asghar	rehan@gmail.com	rehan1234	5004	2026-05-20 00:00:00	Scheduled	8004	Cholesterol
laksh18	053427898	patient	4004	Rehan Asghar	rehan@gmail.com	rehan1234	5004	2026-05-20 00:00:00	Completed	8004	Cholesterol
gki28	3001918781	patient	4005	Talha Ahmed	talha@gmail.com	talha1234	5005	2026-05-06 00:00:00	Completed	8005	Seasonal Flu
gki28	3001918781	patient	4005	Talha Ahmed	talha@gmail.com	talha1234	5005	2026-05-06 00:00:00	Completed	8005	Seasonal Flu
gki28	3001918781	patient	4005	Talha Ahmed	talha@gmail.com	talha1234	5005	2026-05-06 00:00:00	Completed	8005	Seasonal Flu
gki28	3001918781	patient	4005	Talha Ahmed	talha@gmail.com	talha1234	5005	2026-05-06 00:00:00	Completed	8005	Seasonal Flu

W	X	Y	Z	AA	AB	AC	AD
treatment	record_date	log_id	action_performed	log_time	bill_id	amount	payment_status
Paracetamol	2026-03-02 00:00:00	7001	Created	2026-05-02 10:30:00	8001	1500	Unpaid
Paracetamol	2026-03-02 00:00:00	7001	Created	2026-05-02 10:30:00	8001	1500	Unpaid
Lesser salt intake	2026-02-01 00:00:00	7002	Updated	2026-05-02 14:15:00	8002	1500	Unpaid
Lesser salt intake	2026-02-01 00:00:00	7002	Updated	2026-05-02 14:15:00	8002	1500	Unpaid
Lesser salt intake	2026-02-01 00:00:00	7002	Updated	2026-05-02 14:15:00	8002	1500	Unpaid
Lesser salt intake	2/1/2026	7002	Updated	2026-05-02 14:15:00	8002	1500	Unpaid
Biopsy	8/1/2026	7003	Canceled	2026-05-03 09:00:00	8003	1500	Paid
Biopsy	8/1/2026	7003	Canceled	2026-05-03 09:00:00	8003	1500	Paid
Lesser salt intake	3/4/2026	7004	Created	2026-12-25 18:00:00	8004	1500	Unpaid
Lesser salt intake	3/4/2026	7004	Created	2026-12-25 18:00:00	8004	1500	Unpaid
Lesser salt intake	3/4/2026	7004	Created	2026-12-25 18:00:00	8004	1500	Unpaid
Lesser salt intake	3/4/2026	7004	Created	2026-12-25 18:00:00	8004	1500	Unpaid
Lesser salt intake	3/4/2026	7004	Created	2026-12-25 18:00:00	8004	1500	Unpaid
Lesser salt intake	3/4/2026 0:00	7004	Created	2026-12-25 18:00:00	8004	1500	Unpaid
Lesser salt intake	3/4/2026 0:00	7004	Created	2026-12-25 18:00:00	8004	1500	Unpaid
Paracetamol	8/6/2026	7005	Updated	2026-01-01 00:01:00	8005	1500	Paid
Paracetamol	1/1/2026	7005	Updated	2026-01-01 00:01:00	8005	1500	Paid
Panadol	8/6/2026	7005	Updated	2026-01-01 00:01:00	8005	1500	Paid
Panadol	1/1/2026	7005	Updated	2026-01-01 00:01:00	8005	1500	Paid

2NF form

Composite Primary Key: (app_id, patient_id, contact, patient_email, record_id, treatment, record_date, doc_id)

Functional Dependencies:

app_id -> app_date, status
patient_id -> patient_name, patient_password, role
doc_id -> doctor_name, specialization, dept_id
dept_id -> dept_name
admin_id -> username, admin_email, admin_password
record_id -> diagnosis, treatment, record_date
log_id -> action_performed, log_time, app_id
bill_id -> amount, payment_status
(record_id, treatment) -> treatment
(patient_id, contact) -> contact
(patient_id, patient_email) -> email

Partial Dependencies:

Partial Determinant	Dependents
doc_id	doctor_name, specialization
doc_id	dept_id
dept_id	dept_name
patient_id	patient_name, patient_password, role
admin_id	username, admin_password
record_id	diagnosis, record_date
bill_id	amount, payment_status
log_id	action_performed, log_time
app_id	app_date, status

DEPARTMENT TABLE

dept_id	dept_name
1001	Cardiology
1002	Neurology
1003	Psychiatrist
1004	General Physician
1005	Nephrologist

DOCTOR TABLE

doc_id	doctor_name	specialization	dept_id
2001	Dr. Safa Rehman	Cardiologist	1001
2002	Dr. Khadijah Rasheed	Neurosurgeon	1002
2003	Dr. Mohammad Nafees	Psychiatrist	1003
2004	Dr. Mahnoor Mansoor	General Physician	1004
2005	Dr. Maheen Haseeb	Nephrologist	1005

PATIENT TABLE

patient_id	patient_name	patient_password	role
3001	Safwan Alvi	murakami29	patient
3002	Amna Rizvi	habibiuni	patient
3003	Nabiha Ashfaq	polisci23	patient
3004	Laksh Kumar	laksh18	patient
3005	Haider Ali	giki28	patient

PATIENT EMAIL TABLE

patient_id	patient_email
3001	safwan@gmail.com
3002	amnarizvi@example.com
3002	amnarizvi88@example.com
3003	nabihaashfaq@example.com
3004	laksh@example.com
3004	laksh89@example.com
3005	haider@example.com

PATIENT CONTACT TABLE

patient_id	contact
3001	3001234567
3001	34572789
3002	300122697
3002	436789920
3003	3007784967
3004	3002188840
3004	53427898
3005	3001918781

ADMIN TABLE

admin_id	username	admin_email	admin_password
4001	Abdullah Sheikh	abd@gmail.com	abd1234
4002	Plehan Asghar	rehan@gmail.com	rehan1234
4003	Talha Ahmed	talha@gmail.com	talha1234

APPOINTMENT TABLE

app_id	app_date	status	patient_id	doc_id
5001	2026-05-10	Scheduled	3001	2001
5002	2026-05-12	Scheduled	3002	2002
5003	2026-05-15	Completed	3003	2003
5004	2026-05-20	Scheduled	3004	2004
5005	2026-05-06	Completed	3005	2005

MEDICAL RECORD TABLE

record_id	diagnosis	record_date	patient_id	doc_id
6001	Seasonal Flu	2026-03-02	3001	2001
6002	Cholesterol	2026-02-01	3002	2002
6003	Ulcer	2026-08-01	3003	2003
6004	Cholesterol	2026-03-04	3004	2004
6005	Seasonal Flu	2026-08-06	3005	2005

RECORD TREATMENT TABLE

record_id	treatment
6001	Paracetamol
6002	Lesser salt intake
6003	Biopsy
6004	Lesser salt intake
6005	Paracetamol
6005	Panadol

APPOINTMENT LOG TABLE

log_id	action_performed	log_time	app_id
7001	Created	2026-05-02 10:30	5001
7002	Updated	2026-05-02 14:15	5002
7003	Canceled	2026-05-03 09:00	5003
7004	Created	2026-12-25 18:00	5004
7005	Updated	2026-01-01 00:01	5005

BILLING TABLE

bill_id	amount	payment_status	app_id
8001	1500	Unpaid	5001
8002	1500	Unpaid	5002
8003	1500	Paid	5003
8004	1500	Unpaid	5004
8005	1500	Paid	5005

3NF form

Table	Transitive (DNF)	Dependent Attribute	
Doctors	doc_id → dept_id → dept_name	dept_name	
Medical_Records	record_id → patient_id → patient_name	patient_name	300
Appointments	app_id → patient_id → patient_name	patient_name	300
AppointmentLogs	log_id → app_id → app_date	app_date	300
Billing	bill_id → app_id → patient_id	patient_id	300
Departments	dept_id → dept_name	(only one non-key attr)	300
Patients	patient_id → patient_name, password, role	(all directly on PK)	
Admins	admin_id → username, email, password	(all directly on PK)	
DEPARTMENT TABLE			
			300
			300
dept_id	dept_name		300
1001	Cardiology		300
1002	Neurology		300
1003	Psychiatrist		300
1004	General Physician		300
1005	Nephrologist		300
DOCTOR TABLE			P
doc_id	doctor_name	specialization	dept_id
2001	Dr. Safa Rehman	Cardiologist	1001
2002	Dr. Khadijah Rasheed	Neurosurgeon	1002
2003	Dr. Mohammad Nafees	Psychiatrist	1003
2004	Dr. Mahnoor Mansoor	General Physician	1004
2005	Dr. Maheen Haseeb	Nephrologist	1005
			300
			300

RECORD TREATMENT TABLE			
record_id	treatment		
6001	Paracetamol		
6002	Lesser salt intake		
6003	Biopsy		
6004	Lesser salt intake		
6005	Paracetamol		
6006	Panadol		
APPOINTMENT LOG TABLE			
log_id	action_performed	log_time	app_id
7001	Created	2028-05-02 10:30	5001
7002	Updated	2028-05-02 14:15	5002
7003	Canceled	2028-05-03 09:00	5003
7004	Created	2028-12-25 18:00	5004
7005	Updated	2028-01-01 00:01	5005
BILLING TABLE			
bill_id	amount	payment_status	app_id
8001	1500	Unpaid	5001
8002	1500	Unpaid	5002
8003	1500	Paid	5003
8004	1500	Unpaid	5004
8005	1500	Paid	5005

PATIENT TABLE			
patient_id	patient_name	patient_password	role
3001	Safwan Alvi	murakam129	patient
3002	Amna Rizvi	habibiuni	patient
3003	Nabiha Ashfaq	polisci23	patient
3004	Laksh Kumar	laksh18	patient
3005	Haider Ali	gik128	patient
PATIENT EMAIL TABLE			
patient_id	patient_email		
3001	safwan@gmail.com		
3002	amnarizvi@example.com		
3002	amnarizvi88@example.com		
3003	nabihaashfaq@example.com		
3004	laksh@example.com		
3004	laksh89@example.com		
3005	haider@example.com		
PATIENT CONTACT TABLE			
patient_id	contact		
3001	3001234567		
3001	34572789		
3002	300122897		
3002	436788920		
3003	3007784967		
3004	3002188840		

A	B	C	D	E
ADMIN TABLE				
admin_id	username	admin_email	admin_password	
4001	Abdullah Sheikh	abd@gmail.com	abd1234	
4002	Rehan Asghar	rehan@gmail.com	rehan1234	
4003	Taha Ahmed	taha@gmail.com	taha1234	
APPOINTMENT TABLE				
app_id	app_date	status	patient_id	doc_id
5001	2028-05-10	Scheduled	3001	2001
5002	2028-05-12	Scheduled	3002	2002
5003	2028-05-15	Completed	3003	2003
5004	2028-05-20	Scheduled	3004	2004
5005	2028-05-08	Completed	3005	2005
MEDICAL RECORD TABLE				
record_id	diagnosis	record_date	patient_id	doc_id
6001	Seasonal Flu	2028-03-02	3001	2001
6002	Cholesterol	2028-02-01	3002	2002
6003	Ulcer	2028-08-01	3003	2003
6004	Cholesterol	2028-03-04	3004	2004
6005	Seasonal Flu	2028-08-06	3005	2005

BCNF form

A	B	C	D
Table	Functional Dependency	Is Determinant a Superkey?	
Departments	dept_id → dept_name	Yes — dept_id is the sole PK (superkey)	
Doctors	doc_id → doctor_name, specialization, dept_id	Yes — doc_id is the PK (superkey)	
Patients	patient_id → patient_name, password, role	Yes — patient_id is the PK (superkey)	
Patient_Email	(patient_id, patient_email) → record	Yes — composite PK is superkey	
Patient_Contact	(patient_id, contact) → record	Yes — composite PK is superkey	
Admins	admin_id → username, email, password	Yes — admin_id is the PK (superkey)	
Admins	username → admin_id, email, password	Yes — username is a candidate key (UNIQUE NOT NULL) → also a superkey	
Appointments	app_id → app_date, status, patient_id, doc_id	Yes — app_id is the PK (superkey)	
Medical_Records	record_id → diagnosis, record_date, patient_id, doc_id	Yes — record_id is the PK (superkey)	
Record_Treatment	(record_id, treatment) → entry	Yes — composite PK is superkey	
AppointmentLogs	log_id → action_performed, log_time, app_id	Yes — log_id is the PK (superkey)	
Billing	bill_id → amount, payment_status, app_id	Yes — bill_id is the PK (superkey)	
DEPARTMENT TABLE			
dept_id (PK)	dept_name		
1001	Cardiology		
1002	Neurology		
1003	Psychiatrist		
1004	General Physician		
1005	Nephrologist		
DOCTOR TABLE			
doc_id (PK)	doctor_name	specialization	dept_id (FK)
2001	Dr. Safa Rehman	Cardiologist	1001
2002	Dr. Khadijah Rasheed	Neurosurgeon	1002
2003	Dr. Mohammad Nafees	Psychiatrist	1003

ADMIN TABLE				
admin_id (PK)	username	admin_email	admin_password	
4001	Abdullah Sheikh	abd@gmail.com	abd1234	
4002	Rehan Asghar	rehan@gmail.com	rehan1234	
4003	Talha Ahmed	talha@gmail.com	talha1234	
APPOINTMENT TABLE				
app_id (PK)	app_date	status	patient_id (FK)	doc_id (FK)
5001	2026-05-10	Scheduled	3001	2001
5002	2026-05-12	Scheduled	3002	2002
5003	2026-05-15	Completed	3003	2003
5004	2026-05-20	Scheduled	3004	2004
5005	2026-05-06	Completed	3005	2005
MEDICAL RECORD TABLE				
record_id (PK)	diagnosis	record_date	patient_id (FK)	doc_id (FK)
6001	Seasonal Flu	2026-03-02	3001	2001
6002	Cholesterol	2026-02-01	3002	2002
6003	Ulcer	2026-08-01	3003	2003
6004	Cholesterol	2026-03-04	3004	2004
6005	Seasonal Flu	2026-08-06	3005	2005

PATIENT TABLE			
patient_id (PK)	patient_name	patient_password	role
3001	Safwan Ali	murakam29	patient
3002	Amna Rizvi	habibuni	patient
3003	Nabiha Ashfaq	polisc23	patient
3004	Laksh Kumar	laksh18	patient
3005	Haider Ali	gku28	patient
PATIENT EMAIL TABLE			
patient_id (PK, FK)	patient_email (PK)		
3001	safwan@gmail.com		
3002	amnarizvi@example.com		
3002	amnarizv88@example.com		
3003	nabihaashfaq@example.com		
3004	laksh@example.com		
3004	laksh89@example.com		
3005	haider@example.com		
PATIENT CONTACT TABLE			
patient_id (PK,FK)	contact (FK)		
3001	3001234567		
3001	34572789		
3002	300122697		
3002	436788920		
3003	3007794967		
3004	3002188840		

RECORD TREATMENT TABLE			
record_id (PK,FK)	treatment (PK)		
6001	Paracetamol		
6002	Lesser salt intake		
6003	Biopsy		
6004	Lesser salt intake		
6005	Paracetamol		
6005	Panadol		
APPOINTMENT LOG TABLE			
log_id (PK)	action_performed	log_time	app_id (FK)
7001	Created	2026-05-02 10:30	5001
7002	Updated	2026-05-02 14:15	5002
7003	Canceled	2026-05-03 09:00	5003
7004	Created	2026-12-25 18:00	5004
7005	Updated	2026-01-01 00:01	5005
BILLING TABLE			
bill_id (PK)	amount	payment_status	app_id (FK)
8001	1500	Unpaid	5001
8002	1500	Unpaid	5002
8003	1500	Paid	5003
8004	1500	Unpaid	5004
8005	1500	Paid	5005

Problem Analysis

The hospital management in Pakistan is mostly done manually due to which: data is not centralized, the appointments can clash on the same time and for the same doctor, Patients travelled to hospital or contacted on phone to book or cancel an appointment.

The database can solve almost problems stated above and the rest can be solved through the web app. The SQL constraints will validate the information of the doctors and the patients.

Solution Design

The Hospital Appointment Management System is designed as a three-tier web application. The presentation tier consists of html/CSS. The application tier has Python backend that helps routing, and database communication. The data tier has PostgreSQL database that stores all the records.

Functionality & Features

1. Patient

- ❖ **Account Management:** Patients must be able to register for a new account and securely log into the system.
- ❖ **Appointment Booking:** Patients must have the ability to easily book their appointments online without visiting the hospital manually.
- ❖ **Appointment History:** The system must allow patients to view a record of their appointment history.

2. Doctor Functions

- ❖ **Schedule Management:** Doctors must be able to view their schedules within the system.
- ❖ **Appointment History:** The system must allow doctors to view a record of their patient's appointment history.

4. Administrator Functions

- ❖ **Billing Oversight:** The administrator must be able to manage the billing system of patients.
- ❖ **Appointment Oversight:** Administrators must have the authority to manage all hospital appointments.

Implementation and testing phase

Phase 1

Choosing a database was a very difficult task, as the project was to be hosted locally. The appropriate choice was PostgreSQL. The schema was implemented with appropriate constraints and keys. Most importantly the database was firstly normalized into BCNF form so that the data consistency is ensured.

Phase 2

The python backend was planned to be implemented in a single file to keep everything sorted, different functions were defined for every feature of the project. The backend was integrated into the front end through flask app. The python functions validated the execution of the SQL queries through psycopg2 library. As the web application was to be hosted on a local server, so the credentials of the accessing the database were saved in app file directly.

Phase 3

HTML templates were created for each major screen: registration, book appointment, view appointment, doctor schedule, and admin dashboard etc. CSS was used to apply a clean, and consistent style.

Testing

The testing was done using test cases to check whether each functionality is working properly. The integration had many conflicts as the data wasn't flowing from forms to the database correctly but eventually it was solved ensuring smooth flow.

Project Breakdown Structure

WEEK	Work done	Done By
1	Research and Proposal	Safa, Nafees, Maheen
2	Erd diagram	Safa
2	Normalization	Safa, Nafees
3	Schema and SQL queries	Safa, Maheen, Nafees
3	Backend development	Maheen, Safa
4	Front end development	Nafees, Maheen
4	Project Report	Nafees

Results

The screenshot displays the MediCare HMS web application. The browser's address bar shows the URL 127.0.0.1:5000/history. The application's header includes the MediCare HMS logo and a 'RECORDS' button. The main content area is titled 'Appointment Tracking' and features a 'Dashboard' link. Below the title, there is a section for 'All Appointments' which contains a table with the following data:

DATE	PATIENT	DOCTOR	STATUS	PAYMENT	ACTIONS
2026-05-13	Safwan Alvi	Dr. Maria Sheikh	SCHEDULED	UNPAID	Complete Mark Paid

127.0.0.1:5000/history

School

MediCare HMS

RECORDS

OVERVIEW

Appointment Tracking

Dashboard

All Appointments

DATE	PATIENT	DOCTOR	STATUS	PAYMENT	ACTIONS
2026-05-13	Safwan Alvi	Dr. Maria Sheikh	COMPLETED	PAID	Cleared

127.0.0.1:5000

School

MediCare HMS

Appointment #1 marked as Completed.

Bill for Appointment #1 marked as Paid.

Sarah Wahab · Admin

Logout

Access appointments, medical records, and patient management
— all in one secure, unified platform built for modern healthcare.

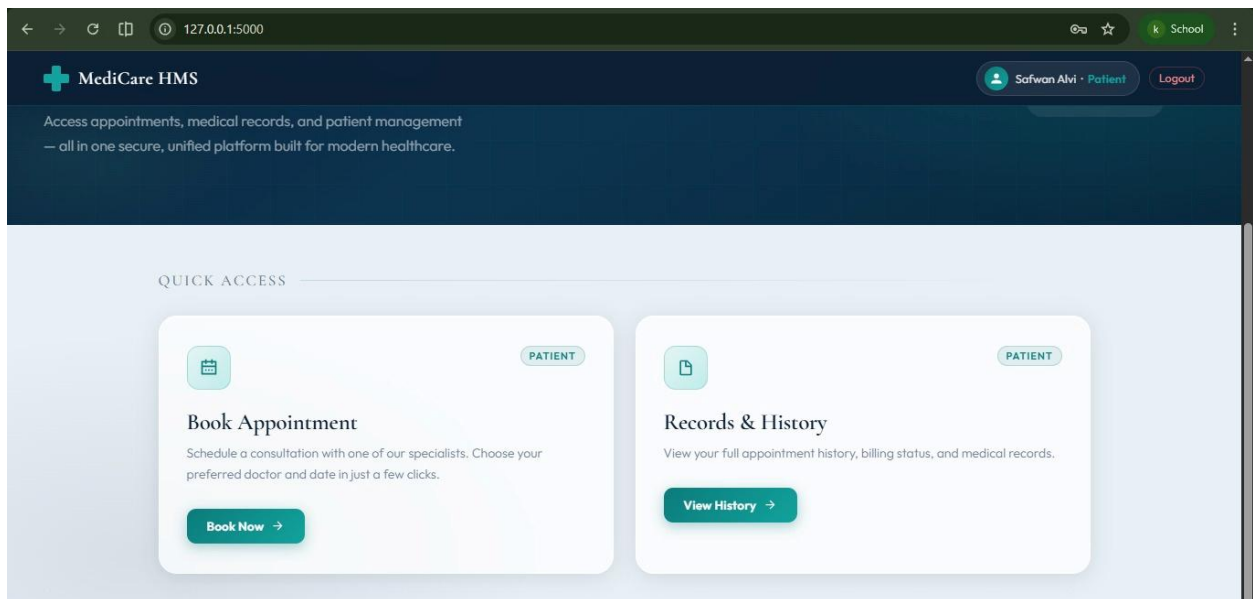
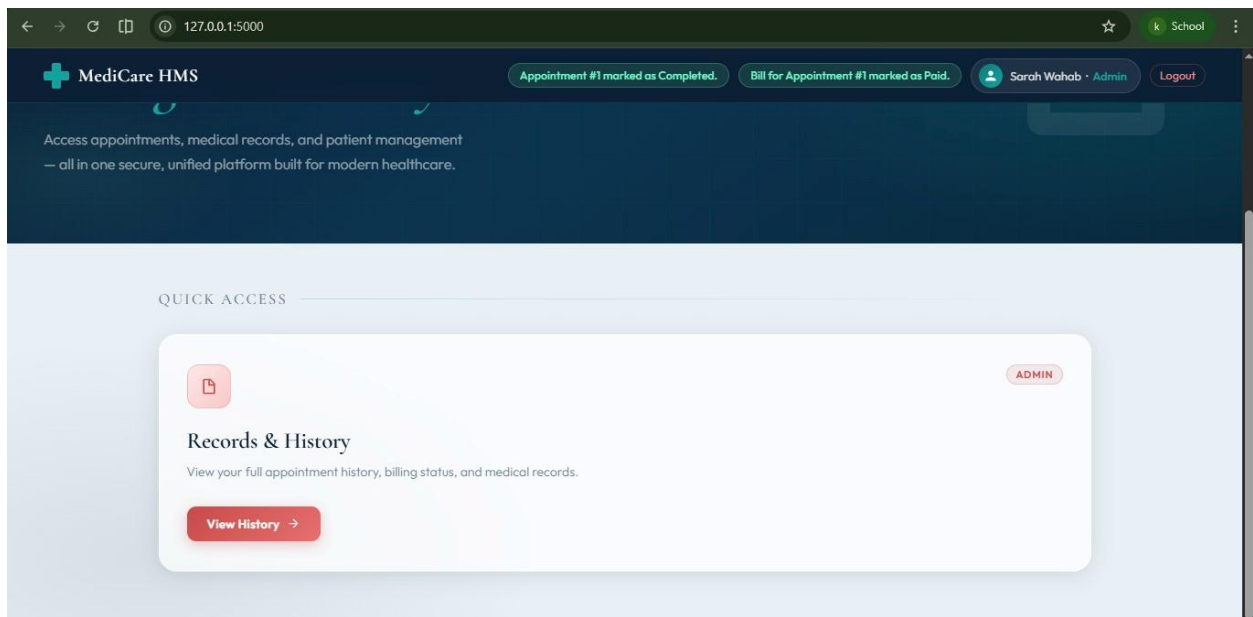
QUICK ACCESS

Records & History

View your full appointment history, billing status, and medical records.

View History →

ADMIN



MediCare HMS

Schedule your *visit* today.

Book a consultation with one of our experienced specialists. Quick, secure, and confirmed instantly.

- 1 Choose a doctor
Select from our verified specialists
- 2 Pick your date
Choose a convenient appointment date
- 3 Confirm booking
Your appointment is instantly confirmed
Secure & HIPAA-compliant platform

NEW APPOINTMENT

Book a Consultation

SELECT DOCTOR

Dr. Khadijah Rasheed

PREFERRED DATE

05/20/2026

Confirm Booking

Back to Dashboard

MediCare HMS

RECORDS

OVERVIEW

Appointment Tracking

Dashboard

All Appointments

DATE	PATIENT	DOCTOR	STATUS	PAYMENT
2026-05-13	Safwan Alvi	Dr. Maria Sheikh	COMPLETED	PAID
2026-05-20	Safwan Alvi	Dr. Khadijah Rasheed	SCHEDULED	UNPAID

Conclusion

The Hospital Appointment Management System successfully achieves the goals set out in the original project proposal. A fully functional web application now allows patients to register, log in, and book appointments with doctors without visiting the hospital in person. Doctors can view and manage their schedules, and administrators have comprehensive oversight. All backed by a well-structured PostgreSQL database. This project demonstrated that even a moderately complex real-world problem, coordinating appointments across

multiple user roles in a healthcare setting can be tackled effectively with the relational database concepts and SQL skills developed during the Database Systems course.